RAJALAKSHMI ENGINEERING COLLEGE

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S2

Started on	Friday, 28 August 2026, 2:13 PM
State	Finished
Completed on	Friday, 28 August 2026, 2:14 PM
Time taken	29 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int floorValue(int a[], int low, int high, int x)
{
    if (low > high)
        return -1;

    int mid = (low + high) / 2;

    if (a[mid] == x)
        return a[mid];

    if (a[mid] < x)
    {
        int f = floorValue(a, mid + 1, high, x);
        return (f == -1) ? a[mid] : f;
    }

    return floorValue(a, low, mid - 1, x);
}

int main()
{
    int n, x, i;
    scanf("%d", &n);

    int a[n];

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    printf("%d", floorValue(a, 0, n - 1, x));

    return 0;
}

```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔

	Input	Expected	Got	
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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